

WASIF ULLAH

Full Stack AI Developer | Python · FastAPI · React · Next.js · Computer Vision

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Summary

Full Stack AI Developer with 2+ years designing and shipping production-grade backend systems, AI integrated web applications, and computer vision pipelines for clients across the UK, EU, and US. Proficient in Python, FastAPI, React.js, Next.js, PostgreSQL, Docker, and AWS delivering end-to-end systems from REST API architecture and database design to responsive frontend and cloud deployment. Proven results: 0.83 mAP on a diagram digitisation pipeline, 97% arrow-association accuracy, and 89% prediction accuracy on a live RAG recommendation system. Experienced working in Agile cross-functional teams delivering for international clients on tight delivery cycles.

Technical Skills

- **Programming Languages:** Python, JavaScript, TypeScript, SQL
- **Backend Development:** FastAPI, Node.js, Express.js, REST APIs, GraphQL, JWT, OAuth, Pydantic, Async Python, Microservices, WebSockets, LiveKit, WebRTC, API Design & Versioning
- **Frontend Development:** React.js, Next.js, Redux Toolkit, Tailwind CSS, shadcn/ui, React Query, Zustand, React Hook Form, Axios, Responsive Web Design
- **Databases:** PostgreSQL, MongoDB, MySQL, Redis, Vector Databases, Pinecone, Schema Design, Query Optimisation, Indexing
- **AI & Computer Vision:** PyTorch, Object Detection, Image Segmentation, OCR, RF-DETR, YOLOv8, SAM2, PaddleOCR, RAG Pipelines, LangChain, Embeddings, Semantic Search, OpenCV
- **Cloud & DevOps:** AWS (EC2, S3, SageMaker, API Gateway, Lambda), Docker, CI/CD Pipelines, GitHub Actions, Terraform, Terragrunt, Git
- **Testing & Quality:** Jest, Pytest, Unit Testing, Integration Testing, Postman, API Testing
- **Tools & Workflow:** Git, GitHub, Agile, Scrum, JIRA, VS Code, Postman

Professional Experience

Full Stack AI Engineer

Jul 2025 – Present

Strateger AI | Lahore, Pakistan (Onsite)

- Architected and deployed the AI Board Scanner an end-to-end diagram recognition system combining RF-DETR object detection, PaddleOCR text extraction, and SAM2 segmentation, achieving 0.83 mAP and 97% arrow-association accuracy against industry benchmarks.
- Built dynamic binary mask pipelines and spatial algorithms using KD-Trees, skeletonisation (BFS), and colour/angle detection logic to reliably extract complex graphs from diagram images with overlapping objects and OCR gaps.
- Developed and dockerised FastAPI REST APIs enabling scalable deployment across AWS EC2 and S3, with structured JSON output consumed by downstream frontend and enterprise integrations.
- Automated cloud infrastructure with Terraform and Terragrunt, supporting consistent rollout of AI microservices across multiple enterprise regions.
- Implemented CI/CD pipelines with GitHub Actions and Docker to streamline build, test, and deployment cycles across the AI backend.
- Collaborated with engineering and product teams to deliver explainable, client-facing AI solutions for UK, EU, and international enterprise clients.

Full Stack Developer

Aug 2024 – Jun 2025

Gojins | Lahore, Pakistan (Onsite)

- Designed and built a comprehensive HRMS managing end-to-end HR processes leave tracking, payroll, attendance, performance reviews, and analytics dashboards backed by secure REST APIs, PostgreSQL, MongoDB, and JWT authentication for a UK enterprise client.
- Developed a real-time Portfolio Builder enabling users to create, update, and publish personal portfolio sites with live editing and custom domain support, built on React.js, Next.js, Redux Toolkit, and Node.js.
- Engineered multiple production web platforms including business management and equipment troubleshooting systems using React.js, Tailwind CSS, Express.js, and dual-database architectures (PostgreSQL + MongoDB).
- Integrated ML-driven business logic for predictive diagnostics, smart recommendations, and adaptive workflow automation across client platforms.
- Maintained code quality and reliability through Jest unit and integration testing; collaborated with cross-functional Agile teams in a UK tech environment to ship features on tight delivery cycles.

Junior Software Developer

Oct 2023 – May 2024

UNAZ Legal Services | Hull, UK (Remote)

- Built a Lawyer Case Management Application backend using Node.js and REST APIs handling document workflows, scheduling, case tracking, and client management for legal professionals.
- Designed and implemented a Zakat Automation System with real-time analytics, PostgreSQL data layer, and cross-module state management via React Context API.
- Applied async programming patterns to optimise notification delivery and case tracking responsiveness across high-volume legal workflows.
- Maintained scalable, modular codebases following best practices in API integration, component architecture, and responsive UI/UX design.

Key Projects

- **AI Board Scanner 0.83 mAP Diagram:** Production pipeline converting diagram images to structured JSON RF-DETR object detection, PaddleOCR text extraction, SAM2 segmentation, and BFS skeletonisation for arrow→shape mapping with 97% association accuracy.
Stack: FastAPI, PyTorch, Docker, AWS EC2/S3, GitHub Actions.
- **AI Career Coach 89% RAG Accuracy:** Job recommendation engine using Retrieval-Augmented Generation Pinecone vector database and embeddings matched candidates to live job-market data achieving 89% predictive accuracy.
Stack: FastAPI, LangChain, Pinecone, Python, REST API.
- **HRMS Platform:** End-to-end Human Resource Management System covering payroll, leave, attendance, performance reviews, and analytics dashboards for a UK enterprise client.
Stack: React.js, Next.js, Redux Toolkit, Node.js, Express.js, PostgreSQL, MongoDB, JWT.
- **Pakistani Sign Language Recognition Final Year Project:** Generative Adversarial Network for real-time image-based sign language recognition to support assistive communication.
Stack: PyTorch, OpenCV, Python, Deep Learning.

Education

BSc (Hons) Computer Science

University of Engineering & Technology, Mardan

GPA: 3.0 / 4.0

Sep 2021 – May 2025

Certifications

- **Computer Vision Specialization** : DeepLearning.AI
- **Applied Machine Learning: Algorithms** : LinkedIn Learning
- **Web Development Specialization** : Vanderbilt University

Oct 2025

Apr 2024

Jan 2023